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Jørgen S. Dokken · Henrik N. Finsberg ·
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The FEniCS Project

The FEniCS 2024 Conference

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Series Foreword

Dear reader,

Scientific research is increasingly interdisciplinary, and both students and experienced researchers often face the need to learn the foundations, tools, and methods of a new research field. This process can be quite demanding, and typically involves extensive literature searches and reading dozens of scientific papers in which the notation and style of presentation varies considerably. Since the establishment of this series in 2016 by founding editor-in-chief Aslak Tveito, the briefs in this series have aimed to ease the process by introducing and explaining important concepts and theories in a relatively narrow field, and to outline open research challenges and pose critical questions on the fundamentals of that field. The goal is to provide the necessary understanding and background knowledge and to motivate further studies of the relevant scientific literature. A typical brief in this series should be around 100 pages and should be well suited as material for a research seminar in a well-defined and limited area of computing.

We publish all items in this series under the SpringerOpen framework, as this allows authors to use the series to publish an initial version of their manuscript that could subsequently evolve into a full-scale book on a broader theme. Since the briefs are freely available online, the authors do not receive any direct income from the sales; however, remuneration is provided for every completed manuscript. Briefs are written on the basis of an invitation from a member of the editorial board. Suggestions for possible topics are most welcome and can be sent to sundnes@simula.no.

March 2023

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Foreword

This book provides highlights of the research presented at the FEniCS 2024 Conference held in Oslo, Norway in June 2024. The selected topics published here show the typical breadth of FEniCS Project-related research as a whole, including new software and algorithm developments, performance assessment of the implementations on advanced computer architectures, and computational simulation of challenging applications. This book not only confirms the original objectives of the FEniCS Project, but it also demonstrates the continuing vitality of the original concepts behind it.

The FEniCS Project was started at an informal meeting that I organized in Chicago in 2003. Since then, FEniCS has grown to an international collaboration involving developers and users at sites across the globe. Starting in 2005, the FEniCS Project has held annual conferences at sites in Europe and the United States, often including tutorial sessions on “how to use it” for beginners. During the COVID-era, the online FEniCS conference utilized the online Gather software that enabled and encouraged personal contact with people you had never met before. I have had the pleasure to attend almost all of the FEniCS conferences, with just a few exceptions.

FEniCS was one of the first software projects to address the automation of computational mathematical modelling. This includes tools to generate key components of scientific simulation software as well as complete end-user systems based on these tools. Users can then write codes to solve challenging problems in fluid dynamics, heat transfer, advanced materials, and many other areas, including what is now known as multiphysics. Novice users have been able to assemble codes for complex simulations using novel models with little help from developers. Initially, such help came primarily via e-mail, but today online forums such as Discourse are used.

One feature of FEniCS was that it encouraged open publication of the algorithms behind the software. This provided academic recognition that is missing from some other software projects. This approach was copied from computer science, where it has long been standard. FEniCS-related research was published in a number of long-standing journals. More recently, dedicated journals devoted to scientific soft-

ware have emerged. Thus, FEniCS pioneered a culture shift regarding how scientific software is viewed and documented.

The FEniCS concept has, since its inception, spurred various forks as people developed different end-user interfaces and internal software implementations to achieve their objectives. One such bifurcation involved the development of a highly parallel version FEniCS-HPC, while the FEniCS Project targeted a broader user interface and internal structure to attract a larger user base. A later bifurcation led to the Firedrake Project. Most recently, a number of forked (DOLFINx, FFCx) and new (Basix) components have been developed within the FEniCS Project itself. Thus, the FEniCS concept has provided an environment in which different approaches were able to flourish, and this aspect continues today.

FEniCS, Firedrake and DUNE rely on the Unified Form Language (UFL) to provide a language to express simulation models based on the variational form of partial differential equations (PDEs). This possibility was already recognized at the end of the first paragraph in Chapter 0 in the first edition of *The Mathematical Theory of Finite Element Methods* co-authored with Susanne C. Brenner. This observation was, a decade later, demonstrated in the software *Analysa*. *Analysa* provided only a limited family of finite elements, including arbitrary-order Lagrange elements, automatically generated by a specific algorithm. One feature of *Analysa* that was ahead of its time was an algebra of domains and corresponding finite element spaces. In *Analysa* it was possible to define a space of functions on the boundary, on the interior, and to define linear functionals and matrices related to these spaces. This provided a way to implement the linear algebra for problem solution precisely. This type of mixed-dimensional functionality only surfaced in FEniCS in the mid-2010s, and is still under active development in the most recent releases of FEniCS and Firedrake.

One valuable aspect of FEniCS and related systems is that they allow algorithm developers a way to make their advances available to a wide audience. A code written from scratch is hard to use and adapt by novices, and often goes underutilized. The FEniCS approach has become quite influential among finite element codes.

FEniCS has allowed the development and testing of new ideas by a very large community, including people with minimal software engineering training. Often a new technique can be tested numerically before expending the effort to understand the new method analytically. For example, the Robin method described in my recent paper with Todd Dupont and Johnny Guzmán was first tested numerically on a simple problem before attempting to show that the method was well posed. In fact, the proof of that took significant time, since it required discovering a new approach to analyzing finite element methods. Without the assurance that the method actually worked in practice, we likely would have abandoned the search for a rigorous explanation of its behaviour.

Today users have many options to choose from when approaching the implementation of a finite element model. This includes Firedrake, NGSolve, and the modern DOLFINx-based version of the FEniCS Project. The legacy version of FEniCS

(DOLFIN) last updated in 2019 is also still widely used. This is not the forum to compare and contrast these different choices, but one can find some guidance from online discussions.

The current book gives a glimpse of the leading edge of FEniCS-related research. It is a good way to find out current research topics and get a sense of directions for the FEniCS Project for the future.

Chicago, USA, 22nd October 2025

L. Ridgway Scott

Preface

The FEniCS Conference 2024 took place between the 12th and 14th June 2024 at the Simula Research Laboratory in Oslo, Norway. It was the 20th — or possibly the 19th — FEniCS Conference, with official records of the 2007 edition remaining elusive, despite some senior community members claiming that they attended. To avoid future confusion, the 2024 edition of the conference will be permanently marked by this first peer-reviewed conference proceedings, published as part of the *Simula SpringerBriefs on Computing* series.

The conference brought together around eighty developers, existing and potential users of the FEniCS ecosystem, as well as mathematicians, computer scientists and application domain specialists interested in numerical methods, their implementation and applications. The talks were held in a single track in the beautifully appointed Hans Petter Langtangen Lecture Hall in the Simula building in central Oslo. It was a fitting venue, honouring a much missed colleague whose contributions to the scientific computing community remain deeply appreciated.

All contributors to the conference were invited to submit a chapter for peer review. From among the forty-nine talks and ten posters presented, twelve contributors expressed their intent to submit a chapter, and nine were accepted following single-blind peer review. These chapters reflect both the diversity and the ongoing vitality of the FEniCS community, highlighting ongoing developments, the broad usage across all areas of scientific study, and the open collaborative spirit that has characterised the project since its inception at the University of Chicago in 2003.

The conference featured the traditional FEniCS competitions, recognising excellence in the categories of:

- *Best poster.* Alena Jarolímová, Charles University, Czech Republic for the poster “‘Determination of Navier’s slip parameter and the inflow velocity using variational data assimilation’”.

- *Best presentation by a doctoral student.* Alexandre Guibert, University of California San Diego, USA, with the presentation “Strongly coupled electrochemical-thermal-fluid models of a battery pack using FEniCS”.
- *Best presentation by a postdoctoral researcher.* Igor Tominec, Stockholm University, Sweden, with the presentation “On the Stokes problem well-posedness under pressure Dirichlet boundary conditions”.
- *Nate Sime’s award for exceptional FEniCS visualisation.* Marc Hirschvogel, Politecnico di Milano, Italy, contained in the presentation “Block Preconditioners for (Moving Domain) Fluid Dynamics Coupled to Physics- and Projection-based Reduced Models”.

Thanks to the generous support of the Ridgway Scott Foundation and Nate Sime, each winner received a cash prize in recognition of their exceptional efforts.

Three travel awards were supported by NumFOCUS, Inc., which has served as fiscal sponsor of the FEniCS Project since 2017, and the poster session by Flax & Teal Ltd.

We would like to thank all authors, reviewers and sponsors for their contributions and engagement, and to the organisers and local hosts for creating such a welcoming environment. Their combined efforts exemplify the openness that continues to drive the FEniCS Project forward.

Oslo, Norway, 22nd October 2025 *Jørgen S. Dokken, Henrik N. T. Finsberg, Jack S. Hale, Marie E. Rognes, Matthew W. Scroggs*

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